Lecture 24

Integration techniques

Uses.

Summary 8: The product, chain and inverse differentiation rules hold on their stated intervals. The derivatives of logarithms and inverse trigonometric functions are known.
Summary 16: Real polynomials divide with remainder and factor into real linear and irreducible quadratic factors, using F. Nonzero polynomials have finitely many roots; real quadratic nonreal roots occur in conjugate pairs.
Summary 19: A linear map on real coordinate columns has a unique standard matrix.
Summary 20: A square coefficient matrix with zero null space is invertible and solves every right side uniquely.
Summary 23: All primitives on an interval differ by a constant. Substitution retains the inner derivative. The product rule gives integration by parts, including its primitive-existence hypotheses.

Objectives. *Integration rules become useful by matching the integrand's structure. Parts handles products, partial fractions separates rational functions, and trigonometric identities or a specified inverse branch handle powers and square roots. Polynomial factorisation and the square-matrix criterion justify the partial-fraction decomposition.*

By the end of this lecture you will be able to:

- Apply parts to polynomial and logarithmic factors and solve a cyclic exponential-trigonometric case.
- Prove partial-fraction existence from the stated factorisation and square-system criteria and compute a decomposition.
- Integrate linear-power and irreducible-quadratic terms, including repeated quadratic powers.
- Choose trigonometric identities or an invertible substitution while retaining the original interval and branch.

24.1 Parts patterns

All indefinite integrals in this lecture refer to primitive families on a specified nonempty open interval. Each final expression includes one constant there, by Lecture 23. For differentiable u and v, finding a primitive H of $u'v$ gives the primitive $uv - H$ of uv'. The choice of factors determines the new integrand.

For $\int xe^x \, dx$ on $\mathbb{R}$, choose $u = x$ and $v' = e^x$. Then $v = e^x$, and the new integrand $u'v = e^x$

has primitive $H = e^x$. Thus

$$\int xe^x \,\mathrm{d}x = (x-1)e^x + C.$$

Its derivative is $e^x + (x-1)e^x = xe^x$. A higher polynomial factor can be reduced by applying the same rule again, with its degree decreasing each time it is differentiated.

For $\int \ln x \,\mathrm{d}x$ on $(0,\infty)$, write the integrand as $(\ln x)\cdot 1$. Choose $u = \ln x$ and $v' = 1$. Then $v = x$ and $u'v = (1/x)x = 1$, whose primitive is x. The result is

$$\int \ln x \,\mathrm{d}x = x\ln x - x + C \qquad (x > 0).$$

The derivative is $\ln x + 1 - 1 = \ln x$. The logarithm's domain is part of the answer.

A cyclic parts calculation returns to the original integrand rather than lowering a polynomial degree. For $e^x \sin x$, choose $u = \sin x$ and $v' = e^x$ first. The new product $u'v = e^x \cos x$ then uses $u = \cos x$ and $v' = e^x$, whose new product is $-e^x \sin x$. The two product identities that these parts steps combine are

$$(e^x \sin x)' = e^x \sin x + e^x \cos x, \qquad (e^x \cos x)' = e^x \cos x - e^x \sin x.$$

Subtracting the second derivative from the first gives $2e^x \sin x$. This constructs a primitive without first assuming that a primitive of either cyclic integrand exists:

$$\int e^x \sin x \,\mathrm{d}x = \frac{e^x(\sin x - \cos x)}{2} + C.$$

The product identities verify its derivative directly on $\mathbb{R}$. The factor 2 comes from collecting the returned integrand; dropping it would give twice the required derivative.

24.2 Why partial fractions exist

A *rational function* is a quotient P/Q of real polynomials with Q not the zero polynomial. Its real domain excludes the zeros of Q. If Q is a nonzero constant, integrate the polynomial P/Q by the power rule. Otherwise divide polynomials as in Lecture 16:

$$P = QS + R, \qquad \frac{P}{Q} = S + \frac{R}{Q},$$

where $R = 0$ or $\deg R < \deg Q$. The polynomial part S is integrated term by term. A quotient with zero numerator or numerator degree less than denominator degree is called *proper*. Only the proper remainder needs partial fractions. Every formula retains the original denominator's domain, even if algebra cancels a factor. Apply the following theorem to R/Q after division; in its statement, P denotes any proper numerator. Dividing both numerator and denominator by the leading coefficient of Q lets us assume that Q is monic.

Factor that denominator using Lecture 16, with its stated dependence on F:

$$Q = \prod_{i=1}^{s} p_i^{m_i}.$$

Here the p_i are distinct monic real linear or irreducible quadratic polynomials, and $m_i \geq 1$ are integers. For a linear factor the numerator in a partial fraction is a constant. For a quadratic factor it is a polynomial $Ax + B$ of degree at most 1, including zero choices of either coefficient. Every power from 1 to m_i must be available, even if some resulting numerators turn out to be zero.

Theorem 24.1 (Partial fractions). *Let P, Q be real polynomials with Q monic and nonconstant, and with $P = 0$ or $\deg P < \deg Q$. Suppose $Q = \prod_{i=1}^{s} p_i^{m_i}$, where the p_i are distinct monic real linear or irreducible quadratic polynomials and the $m_i \geq 1$ are integers. There are unique real numerator polynomials N_{ik}, constant for linear p_i and of degree at most 1 for quadratic p_i, such that*

$$\frac{P(x)}{Q(x)} = \sum_{i=1}^{s} \sum_{k=1}^{m_i} \frac{N_{ik}(x)}{p_i(x)^k} \qquad \text{whenever } Q(x) \neq 0.$$

Proof. Let $N = \deg Q = \sum_i m_i \deg p_i$. The proposed numerators contain exactly N real scalar coefficients. Clear the denominator. Each term $N_{ik}Q/p_i^k$ has degree less than N when nonzero. Recording its N coefficients, with zeros in unused high positions, makes the cleared numerator a linear function of the N unknown coefficients. Thus coefficient matching is a square real linear system.

We prove that its homogeneous system has only the zero solution. Fix a factor p_i and choose a complex root α of that factor. For a linear factor this is its real root. For an irreducible quadratic it is a nonreal root supplied by complex factorisation in Lecture 16. No other factor vanishes at α. Distinct linear factors have different roots, and irreducible quadratics have no real root. A real linear factor cannot vanish at a nonreal point. If two monic real quadratics shared a nonreal root, conjugation would give both roots in common and the quadratics would be the same, contrary to distinctness. Consequently the polynomial $H_i = Q/p_i^{m_i}$ satisfies $H_i(\alpha) \neq 0$.

In the cleared homogeneous identity, all terms except $N_{i,m_i}H_i$ contain a factor p_i. Evaluation at α gives $N_{i,m_i}(\alpha) = 0$. For a linear factor this numerator is constant, so it is zero. For a quadratic factor, write the numerator as $N_{i,m_i}(x) = Ax + B$, with $A, B \in \mathbb{R}$. The identity at the root is $A\alpha + B = 0$. Taking the imaginary part of this identity gives $A \operatorname{Im} \alpha = 0$. Since α is nonreal, $\operatorname{Im} \alpha \neq 0$, so $A = 0$. The identity then gives $B = 0$.

If $m_i > 1$, remove this zero numerator and divide the remaining polynomial identity by p_i; every remaining term is divisible by it. The same evaluation now forces $N_{i,m_i-1} = 0$. Continue while a power remains, until every numerator associated with this factor is zero. At each step the other-factor terms still contain p_i until this list is exhausted. If $m_i = 1$, the first evaluation has already finished this factor. Repeat from the original identity for each factor. All unknown coefficients are zero, as claimed.

The square-system criterion of Lecture 20 therefore makes the coefficient map invertible. It gives one and only one set of numerators for every proper numerator P. This constructs a polynomial identity and hence the required quotient identity on its domain. Conversely, any such quotient identity clears to a polynomial vanishing at every real point where $Q \neq 0$. That is an infinite set, while a nonzero polynomial has finitely many roots by Lecture 16. Thus it is the same polynomial identity, and the proved uniqueness of its coefficients applies. ∎

Example 24.2 (Repeated linear and quadratic factors). Decompose and integrate

$$\frac{3x^3 - 3x^2 - 5x - 1}{(x-1)^2(x^2+1)} \qquad (x \neq 1).$$

The proper quotient requires the form

$$\frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{Lx + M}{x^2+1}.$$

Clearing the denominator gives

$$3x^3 - 3x^2 - 5x - 1 = A(x-1)(x^2+1) + B(x^2+1) + (Lx+M)(x-1)^2.$$

Matching the coefficients of $x^3, x^2, x, 1$, with unknowns ordered as A, B, L, M, gives the augmented coefficient system

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 3 \\ -1 & 1 & -2 & 1 & -3 \\ 1 & 0 & 1 & -2 & -5 \\ -1 & 1 & 0 & 1 & -1 \end{array}\right).$$

Add row 1 to rows 2 and 4, and subtract row 1 from row 3. Then subtract row 2 from row 4 and exchange rows 3 and 4. The echelon array is

$$\left(\begin{array}{cccc|c} 1 & 0 & 1 & 0 & 3 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & -2 & -8 \end{array}\right).$$

Back substitution gives $M = 4$, $L = 1$, $B = L - M = -3$ and $A = 3 - L = 2$. The decomposition and primitive are therefore

$$\frac{3x^3 - 3x^2 - 5x - 1}{(x-1)^2(x^2+1)} = \frac{2}{x-1} - \frac{3}{(x-1)^2} + \frac{x+4}{x^2+1},$$

$$\int \frac{3x^3 - 3x^2 - 5x - 1}{(x-1)^2(x^2+1)}\,dx = 2\ln|x-1| + \frac{3}{x-1} + \frac{1}{2}\ln(x^2+1) + 4\arctan x + C.$$

The first two derivatives give the two linear-factor terms. The final two derivatives combine to $(x+4)/(x^2+1)$, the quadratic-factor term. The decomposition was proved by its coefficient identity, so this checks the original integrand as well. There is one independent constant on $(-\infty, 1)$ and another on $(1, \infty)$. The quadratic factor is positive and introduces no extra real domain break.

Example 24.3 (Polynomial division and distinct real poles)**.** Integrate the improper rational function

$$\frac{x^3 + 2x^2 - x + 4}{(x+1)(x-3)} \qquad (x \neq -1, 3).$$

Polynomial division gives the identity

$$x^3 + 2x^2 - x + 4 = (x+4)(x^2 - 2x - 3) + (10x + 16).$$

For the proper remainder, write

$$\frac{10x + 16}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}.$$

Clearing the denominator gives $A + B = 10$ and $-3A + B = 16$. Subtracting these equations gives $-4A = 6$, so $A = -3/2$ and $B = 23/2$. The polynomial part and two logarithmic parts yield

$$\int \frac{x^3 + 2x^2 - x + 4}{(x+1)(x-3)}\,dx = \frac{x^2}{2} + 4x - \frac{3}{2}\ln|x+1| + \frac{23}{2}\ln|x-3| + C.$$

Differentiation returns $x + 4 - 3/(2(x+1)) + 23/(2(x-3))$, which is the original quotient by the two polynomial identities above. The formula has independent constants on $(-\infty, -1)$, $(-1, 3)$ and $(3, \infty)$. Each excluded endpoint is a genuine pole for these data.

24.3 The rational model terms

For a linear factor with real root a, the power and logarithm rules give

$$\int \frac{\mathrm{d}x}{(x-a)^k} = \begin{cases} \ln|x-a| + C, & k = 1, \\ \dfrac{(x-a)^{1-k}}{1-k} + C, & k \geq 2, \end{cases}$$

where k is a positive integer and the interval excludes a. The derivative of $x - a$ is 1, so there is no additional scale factor.

For an irreducible quadratic, write $D(x) = x^2 + px + q$ with $p, q \in \mathbb{R}$ and $4q - p^2 > 0$. Completing the square gives

$$D(x) = (x + p/2)^2 + \delta, \qquad \delta = q - p^2/4 > 0.$$

Split a numerator by its derivative term: $Ax + B = (A/2)D'(x) + (B - Ap/2)$. Substitution in the first term and the arctangent row in the second give

$$\int \frac{Ax+B}{D(x)}\,\mathrm{d}x = \frac{A}{2}\ln D(x) + \frac{B - Ap/2}{\sqrt{\delta}}\arctan\left(\frac{x + p/2}{\sqrt{\delta}}\right) + C. \tag{24.1}$$

Indeed, setting $t = (x + p/2)/\sqrt{\delta}$ gives $\mathrm{d}x = \sqrt{\delta}\,\mathrm{d}t$ and $D(x) = \delta(1 + t^2)$. Thus a primitive of $1/D(x)$ is $\arctan((x+p/2)/\sqrt{\delta})/\sqrt{\delta}$. The quadratic is positive on all of $\mathbb{R}$, so no absolute-value branch or missing real point is needed. For example, $3x - 2 = \frac{3}{2}(2x + 4) - 8$ gives

$$\int \frac{3x-2}{x^2+4x+7}\,\mathrm{d}x = \frac{3}{2}\ln(x^2+4x+7) - \frac{8}{\sqrt{3}}\arctan\left(\frac{x+2}{\sqrt{3}}\right) + C.$$

Differentiating the logarithm contributes $(3x + 6)/D(x)$ and differentiating the arctangent term contributes $-8/D(x)$, giving the required numerator.

For a repeated quadratic power D^k, use the same numerator split. The derivative term integrates by the power rule when $k \geq 2$:

$$\int \frac{D'(x)}{D(x)^k}\,\mathrm{d}x = \frac{D(x)^{1-k}}{1-k} + C.$$

The remaining constant numerator reduces under the same variable t to $\delta^{1/2-k}/(1 + t^2)^k$. These last model terms have primitives constructed recursively. Set $J_1(t) = \arctan t$. For integers $k \geq 2$, the product rule gives

$$\left(\frac{t}{(1+t^2)^{k-1}}\right)' = \frac{2k-2}{(1+t^2)^k} - \frac{2k-3}{(1+t^2)^{k-1}}.$$

Consequently the recurrence

$$J_k(t) = \frac{t}{2(k-1)(1+t^2)^{k-1}} + \frac{2k-3}{2(k-1)}J_{k-1}(t)$$

constructs a primitive of $1/(1 + t^2)^k$. Differentiating the two terms cancels the power-$k - 1$ contribution and leaves precisely the power-k integrand. Starting from J_1 proves existence for every positive integer k. For instance, $J_2 = t/(2(1 + t^2)) + \arctan t/2$. Thus every term in the proved partial-fraction decomposition has a primitive obtained from the displayed rules.

24.4 Trigonometric powers and a specified branch

For nonnegative integer powers of sine and cosine, an odd power supplies one derivative factor for substitution. For example, in $\sin^3 x \cos^2 x$, keep one $\sin x$ and replace $\sin^2 x$ by $1 - \cos^2 x$. The substitution $u = \cos x$ gives

$$\int \sin^3 x \cos^2 x \, dx = -\int (1 - u^2) u^2 \, du = \frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C.$$

Its derivative is $\sin x \cos^2 x (1 - \cos^2 x) = \sin^3 x \cos^2 x$. This formula holds on $\mathbb{R}$, including zeros of the derivative factor. If instead a cosine power is odd, keeping one cosine supplies the substitution by $\sin x$ in the same way. When both powers are even, the angle-addition identities lower powers. For example,

$$\sin^2 x \cos^2 x = \frac{\sin^2(2x)}{4} = \frac{1 - \cos 4x}{8}$$

gives

$$\int \sin^2 x \cos^2 x \, dx = \frac{x}{8} - \frac{\sin 4x}{32} + C.$$

Differentiation restores $(1 - \cos 4x)/8$, so this answer is also valid on $\mathbb{R}$.

A substitution for a square root requires an explicit inverse branch. To integrate $\sqrt{9 - x^2}$ on $I = (-3, 3)$, take

$$x = 3 \sin t, \qquad -\pi/2 < t < \pi/2, \qquad t = \arcsin(x/3).$$

The sine is one-to-one on this branch and $\cos t > 0$. Thus $\sqrt{9 - x^2} = 3 \cos t$, rather than an unexamined signed square root.

Figure 24.1 shows the positive-x triangle for this branch.

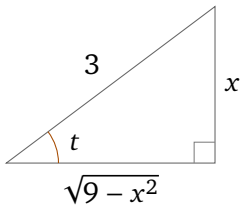


Figure 24.1: For $0 < x < 3$, the right triangle gives $\sin t = x/3$ and $\cos t = \sqrt{9-x^2}/3$. The chosen branch also covers negative x with negative t, while its cosine remains positive.

Since $\mathrm{d}x = 3\cos t\,\mathrm{d}t$, the transformed integrand is $9\cos^2 t$. Its primitive is $9t/2+9\sin(2t)/4$ by the even-power identity. Returning to x uses $\sin(2t) = 2\sin t\cos t = 2x\sqrt{9-x^2}/9$. The resulting family is

$$\int \sqrt{9-x^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{9-x^2} + \frac{9}{2}\arcsin(x/3) + C \qquad (-3 < x < 3).$$

For a derivative check, the two terms contribute $\sqrt{9-x^2}/2 - x^2/(2\sqrt{9-x^2})$ and $9/(2\sqrt{9-x^2})$. Their sum is $\sqrt{9-x^2}$ throughout I. Returning to x applies Theorem 23.2 with the differentiable inverse branch $t = \arcsin(x/3)$ as its inner function; Lecture 8 gives that branch's derivative. The explicit check above also verifies the result. No area interpretation has been assumed.

Summary 24.

- Parts differentiates one selected factor and integrates the other. Repeated polynomial reduction and cyclic product identities give the stated patterns.
- Theorem 24.1: after polynomial division and real factorisation, every proper rational function has the stated unique partial fractions. The proof uses F through factorisation and the square coefficient-system criterion.
- Equation (24.1): linear powers and completed-square quadratic terms integrate by the logarithm, power and arctangent rules. The proved recurrence supplies every repeated quadratic power.
- Odd trigonometric powers can supply a substitution factor; even powers use the addition identities. A radical substitution retains its inverse branch and the sign of the square root.
- All formulas describe primitive families on their original intervals. Differentiate the final answer and retain independent constants across actual domain breaks.

Sheet 24 practices parts, rational and trigonometric methods, including algebraic and tangent-half-angle substitutions. Lecture 25 defines the definite integral through finite lower and upper sums.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§7.1–7.5. V. A. Zorich, *Mathematical Analysis I*, 2nd ed., Springer, 2015, §5.7. The partial-fraction decomposition is stated in Stewart, §7.4; the proof above uses the factorisation and linear-system results of this course.